

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q21-1862

Pine Hollow 138 kV
October 17, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1862. The Interconnection Customer proposes a 255.0 MW solar photovoltaic facility interconnecting at the Pine Hollow 138 kV substation in the Virginia area. The request was received on August 19, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 423 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 6 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$29,913,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
CHARLOTTESVILLE / PINE HOLLOW 138kV	ALBRED A TANS MTN TAP / COLGATE 138kV	165.8	302	100.0
GROVE ST / CHARLOTTESVILLE 138kV	ALBRED A TANS MTN TAP / COLGATE 138kV	126.3	240	56.2
GORDONSVILLE SUBSTATION / PROFFIT 138kV	CHARLOTTESVILLE / PINE HOLLOW 138kV line	119.8	364	100.0
PROFFIT / COLGATE 138kV	CHARLOTTESVILLE SUBSTATION / PINE HOLLOW 138kV L	118.8	385	100.0
COLGATE / ALBRED A TANS MTN TAP SUBSTATION 138kV	PINE HOLLOW / CHARLOTTESVILLE SUBSTATION 138kV	116.9	429	100.0
PENINSULA / PINE HOLLOW 138kV	PINE HOLLOW / CHARLOTTESVILLE 138kV	113.0	240	100.0

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
CHARLOTTESVILLE / PINE HOLLOW 138kV	Rebuild line with higher-capacity conductor	\$7,992,000	52% (\$4,153,000)
GROVE ST SUBSTATION / CHARLOTTESVILLE 138kV	Rebuild line with higher-capacity conductor	\$13,417,000	49% (\$6,573,000)
GORDONSVILLE / PROFFIT SUBSTATION 138kV	Rebuild line with higher-capacity conductor	\$8,598,000	52% (\$4,468,000)
COLGATE / PROFFIT 138kV line	Rebuild line with higher-capacity conductor	\$11,444,000	52% (\$5,946,000)
COLGATE / ALBREDAS MTN TAP SUBSTATION 138kV	Rebuild line with higher-capacity conductor	\$6,128,000	52% (\$3,184,000)
PENINSULA / PINE HOLLOW 138kV line	Rebuild line with higher-capacity conductor	\$5,589,000	100% (\$5,589,000)
	Total allocated to project		\$29,913,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.